REVIEW EX

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Review Exercise I

1). Select The Correct Option.

(i) all of These.

(ii) $|z_1+z_2| \le |z_1| + |z_2|$

(iii) $-3i-4+3i+4 = -8$

(iv) $ab<0$ $(+ve)(-ve) = -ve$

(v) 1 or -1

(vi) x

(vii) $(-1)^{\frac{-21}{2}} = \dfrac{1}{(-1)^{\frac{+21}{2}}} = \dfrac{1}{\left((-1)^{1/2}\right)^{21}}$

$= \dfrac{1}{i^{21}} = \dfrac{1}{i^{20}\,i} = \dfrac{1}{i(i^2)^{10}}$

$= \dfrac{1}{i} = \dfrac{1}{i} \times \dfrac{+i}{+i} = \dfrac{i}{-1}$

$= -i$

(viii) $z = 0+i$, $z^2 = (+i)^2 = -1$

(ix) $z = a+ib$ $\bar{z} = a-ib$

$z\bar{z} = (a+ib)(a-ib) = a^2+b^2$

Real

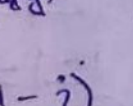
(x) $\Rightarrow \dfrac{2+i}{i} = \dfrac{2+i}{i}\cdot\dfrac{i}{i} = \dfrac{2i+i^2}{-1}$

$= \dfrac{-1+2i}{-1} = 1-2i$

Re: 1

(xi) Complex plane

(xii) origin



(xiii) $z_1z_2 = (2+3i)(1-i)$

$\Rightarrow z_1z_2 = 2-2i+3i-3i^2$

$\Rightarrow z_1z_2 = 2+i+3$

$\Rightarrow z_1z_2 = 5+i$

$|z_1z_2| = \sqrt{5^2+1^2} = \sqrt{26}$

(xiv) $|x+5i| = 3$

$\sqrt{x^2+5^2} = 3$ None

$x^2+25 = 9$ of These

$x^2 = -25+9 = -16$

(xv) $(3,4)$.

$= \left(\dfrac{3}{3^2+4^2}, \dfrac{-4}{3^2+4^2}\right) = \left(\dfrac{3}{25}, \dfrac{-4}{25}\right)$

(xvi) $z = -3i+4$

$\bar{z} = +3i+4$

(xvii) $i(3-2i) = 3i-2i^2 = 3i+2$

Re: 2 Img 3, $\boxed{2 \,\&\, 3}$

(xviii) if $z = a+ib$ $|z| = \sqrt{a^2+b^2}$

(xix) $\left(\dfrac{1+i}{1-i}\right)^{12} = \left(\left(\dfrac{1+i}{1-i}\right)^2\right)^6$

$\Rightarrow \left(\dfrac{1+2i+i^2}{1-2i+i^2}\right)^6 = \left(\dfrac{1+2i-1}{1-2i-1}\right)^6$

$= \left(\dfrac{2i}{-2i}\right)^6 = (-1)^6 = 1$

(xx) $i^{-7} = \dfrac{1}{i^7} = \dfrac{1}{i\,i^6} = \dfrac{1}{i(i^2)^3}$

$= \dfrac{1}{i(-1)^3} = \dfrac{1}{-i} = \dfrac{1}{-i} \times \dfrac{i}{i} = i$

Q#2). Simplify:

(i) $(5-6i)+(5i)+(7+6i)$

$= 12+5i$

(ii) $(4-i)+(-9+6i)$

$= -5+5i$

(iii) $(9+11i)-(3+5i)$

$= 6+6i$

(iv) $(-2-15i)-(-12+13i)$

$= -2-15i+12-13i$

$= 10-28i$

(v) $(-3+2i)(3-8i)$

$= -9+24i+6i-16i^2$ $\because i^2=-1$

$= 7+30i$

(vi) $(3-i)(4+3i)(5-2i)$

$=(3-i)(20-8i+15i-6i^2)$

$=(3-i)(26+7i)$

$=78+21i-26i-7i^2 \quad \because i^2=-1$

$=85-5i$

(vii) $\dfrac{4-i}{6-3i}$

$=\dfrac{4-i}{6-3i}\times\dfrac{6+3i}{6+3i}$

$=\dfrac{24+12i-6i-3i^2}{36-9i^2}$

$=\dfrac{27+6i}{45}=\dfrac{27}{45}+\dfrac{6}{45}i$

$=\dfrac{3}{5}+\dfrac{2}{15}i$

(viii) $\dfrac{8+6i}{6-2i}$

$=\dfrac{8+6i}{6-2i}\times\dfrac{6+2i}{6+2i}$

$=\dfrac{48+16i+36i+12i^2}{36+4}$

$=\dfrac{36+52i}{40}=\dfrac{9}{10}+\dfrac{13}{10}i$

Q#3 (a) Find each of the following

(i) $\bar{z}$ $\quad z=3-15i$

$\bar{z}=3+15i$

(ii) $\overline{z_1+z_2}$

$=\overline{5+i-(8+3i)}=\overline{5+i+8-3i}$

$=\overline{+13+2i}=+3-2i$

(iii) $\overline{z_1 z_2}$ $\quad z_1=-10+5i$

$z_2=5-10i$

$=\overline{(-10+5i).(5-10i)}$

$=\overline{-50+100i+25i-50i^2}$

$=\overline{0+125i}$

$=-125i$

(iv) $\dfrac{z}{\bar{z}}=\dfrac{15-3i}{\overline{15-3i}}=\dfrac{15-3i}{15+3i}$

$\dfrac{z}{\bar{z}}=\dfrac{15-3i}{15+3i}\times\dfrac{15-3i}{15-3i}$

$\dfrac{z}{\bar{z}}=\dfrac{(15-3i)^2}{234}=\dfrac{216-90i}{234}$

$\dfrac{z}{\bar{z}}=\dfrac{12}{13}-\dfrac{5}{13}i$

(v) $\bar{z}_1-\bar{z}_2$

$=\overline{5+4i}-(\overline{-8+5i})=5-4i-(-8-5i)$

$=5-4i+8+5i=13+i$

Q#3 (b) Find absolute value of the following complex number.

(i) $z=-3-6i$

$|z|=\sqrt{(-3)^2+(-6)^2}=\sqrt{9+36}=\sqrt{45}$

$|z|=3\sqrt{5}$

(ii) $z=-6+3i$

$|z|=\sqrt{(-6)^2+(3)^2}=\sqrt{36+9}=3\sqrt{5}$

(iii) $z=\dfrac{6+3i}{10+2i}$

$=\dfrac{6+3i}{10+2i}\cdot\dfrac{10-2i}{10-2i}$

$=\dfrac{60-12i+30i-6i^2}{100+4}$

$=\dfrac{66}{104}-\dfrac{18i}{104}=\dfrac{33}{52}-\dfrac{9}{52}i$

$|z| = \sqrt{\left(\frac{33}{52}\right)^2 + \left(\frac{-9}{52}\right)^2}$

$|z| = \sqrt{\frac{1089}{2704} + \frac{81}{2704}}$

$|z| = \frac{3\sqrt{130}}{52}$

(iv) $\frac{1+i}{1-i}$

$z = \frac{1+i}{1-i} \times \frac{1+i}{1+i} = \frac{1+2i+i^2}{1^2-i^2}$

$z = \frac{2i}{2} = i$

$|z| = \sqrt{0^2+1^2} = \sqrt{1} = 1$

(i)#4). Find the values of x and y if:

(i) $5x+3iy = -x+2iy$

$\Rightarrow 5x+x+3iy-2iy=0$

$\Rightarrow 6x+iy=0$

$x=0, \; y=0$

(ii) $x^2-7x+9iy = iy^2+20i-12$

$\Rightarrow x^2-7x+12+9iy-iy^2+20i=0+0i$

$\Rightarrow (x^2-7x+12)+(-y^2+9y+20)i=0+0i$

$x^2-7x+12=0, \quad -y^2+9y+20=0$

$x^2-4x-3x+12=0, \quad +y^2-9y-20=0$

$x(x-4)-3(x-4)=0 \quad y^2-5y-4y-20=0$

$(x-4)(x-3)=0 \quad y(y-5)-4(y-5)=0$

$x=4, \; x=3 \quad (y-5)(y-4)=0$

$y=5, \; y=4$

$x=3, \; x=4, \; y=4, \; y=5$

(iii)#5 (i) Find real and Imaginary parts of the following.

(i) $z=-3+i-42i$

$z=-3-41i$ Re -3 Img -41

(ii) $\frac{3z+i}{z+4}$ for $z=3+2i$

$= \frac{3(3+2i)+i}{3+2i+4} = \frac{9+6i+i}{7+2i}$

$= \frac{9+7i}{7+2i} \times \frac{7-2i}{7-2i} = \frac{63-18i+49i-14i^2}{49+4}$

$= \frac{77+31i}{53}$

(6) Find the additive and multiplicative inverse of each of the following:

(i) $3-7i$

Additive Inverse $= -3+7i$

Multiplicative Inverse

$\frac{1}{3-7i} = \frac{1}{3-7i} \times \frac{3+7i}{3+7i} = \frac{3+7i}{9+49}$

$= \frac{3+7i}{58} = \frac{3}{58}+\frac{7}{58}i$

(ii) $-2+i$

Additive Inverse $= 2-i$

Multiplicative Inverse $= \frac{1}{-2+i}$

$\frac{1}{-2+i} \times \frac{-2-i}{-2-i} = \frac{-2-i}{4+1}$

$\Rightarrow -\frac{2}{5}-\frac{i}{5}$

(iii) $-2-5i$

Additive Inverse $= 2+5i$

Multiplicative Inverse

$\frac{1}{-2-5i}$

$\frac{1}{-2-5i} \times \frac{-2+5i}{-2+5i} = \frac{-2+5i}{4+25}$

$\Rightarrow \frac{-2+5i}{29}$

(iv) $\frac{1}{2}+\frac{3}{2}i$

Additive Inverse $= -\frac{1}{2}-\frac{3}{2}i$

Multiplicative Inverse $\frac{1}{+\frac{1}{2}+\frac{3}{2}i}$

$= \frac{1}{\frac{1}{2}(1+3i)} = \frac{1}{\frac{1}{2}(1+3i)} \times \frac{1-3i}{1-3i}$

$\Rightarrow \frac{1-3i}{\frac{1}{2}(1+9)} = \frac{1-3i}{5}$

(v) $-4+\sqrt{7}i$

Additive Inverse $= 4-\sqrt{7}i$

Multiplicative Inverse $= \frac{1}{-4+\sqrt{7}i}$

$\Rightarrow \frac{1}{-4+\sqrt{7}i} \times \frac{-4-\sqrt{7}i}{-4-\sqrt{7}i} = -\frac{4-\sqrt{7}i}{16+7}$

$\Rightarrow \frac{-4+\sqrt{7}i}{23} = -\frac{4-\sqrt{7}i}{23}$

(vi) $3+4i$

Additive Inverse $= -3-4i$

Multiplicative Inverse

$\Rightarrow \frac{1}{+3+4i} = \frac{1}{+3+4i} \times \frac{+3+4i}{+3+4i}$

$\Rightarrow \frac{+3+4i}{9+16} = +\frac{3+4i}{25} = \frac{3-4i}{25}$

Q#7). Solve the following equation.

$z^2-(2-3i)z-5+i=0$

$z = \frac{-(-(2-3i)) \pm \sqrt{(2-3i)^2-4(1)(-5+i)}}{2(1)}$

$z = \frac{(2+3i) \pm \sqrt{4-12i+9i^2+20-4i}}{2(1)}$

$z = \frac{(2-3i) \pm \sqrt{24-16i-9}}{2}$

$z = \frac{(2-3i)}{2} \pm \frac{\sqrt{15-16i}}{2}$

Q#8) Verify the following

(i) $(2+3i)^2 = -5+12i$

L.H.S

$4+12i+9i^2 = -5+12i$

(ii) $2(5-2i)^2 = 42-40i$

$= 2(25-20i+4i^2) = 2(25-4-20i)$

$\Rightarrow 2(21-20i) = 42-40i$

(iii) $z^3+i = (z-i)(z^2+iz)-1)$

$= (z-i)(z^2+iz-1)$

$= z^3+iz^2-z-iz^2-i^2z+i$

$= z^3-z+z+i = z^3+i$

Q#9) Solve the quadratic equation by completing the squares.

(i) $z^2-5z+7=0$

$\Rightarrow z^2-5z+\left(\frac{5}{2}\right)^2 = -7+\left(\frac{5}{2}\right)^2$

$\Rightarrow \left(z-\frac{5}{2}\right)^2 = -7+\frac{25}{4} = \frac{-28+25}{4} = -\frac{3}{4}$

$z-\frac{5}{2} = \frac{\pm\sqrt{3}}{2}i \Rightarrow z = \frac{5}{2} \pm \frac{\sqrt{3}}{2}$

(ii) $2z^2-6z+9=0$

$z^2-3z+\frac{9}{2}=0$

$\Rightarrow z^2-3z+\left(\frac{3}{2}\right)^2 = -\frac{9}{2}+\left(\frac{3}{2}\right)^2$

$\Rightarrow \left(z-\frac{3}{2}\right)^2 = \frac{-18+9}{4} = -\frac{9}{4}$

$z-\frac{3}{2} = \pm\frac{3i}{2}$

$z = \frac{3}{2} \pm \frac{3i}{2}$

(iii) $4z^2+25=0$

$z^2+\frac{25}{4}=0 \Rightarrow z^2 = -\frac{25}{4}$

$(z+0)^2 = -\frac{25}{4} \Rightarrow z+0 = \pm\frac{5}{2}i$

$z = \pm\frac{5}{2}i$

(iv) $z^2-10z+41=0$

$\Rightarrow z^2-10z+5^2 = -41+5^2$

$(z-5)^2 = -41+25 = -16$

$z-5 = \pm 4i \Rightarrow z = 5 \pm 4i$

(v) $4z^2+4z+5=0$

$z^2+z+\left(\frac{1}{2}\right) = -\frac{5}{4}+\left(\frac{1}{2}\right)^2 = -1$

$\left(z+\frac{1}{2}\right)^2 = i \Rightarrow z = \frac{1}{2} \pm i = \frac{1 \pm 2i}{2}$